SMIC

Semiconductor Manufacturing International Corporation

SMIC CONFIDENTIAL

Quick-start on SMIC Calibre Connectivity Interface (CCI) - StarRC Flow

Post-layout Extraction Team

September 2015

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Before You Start to Run StarRC by CCI

- SMIC recommends user to execute parasitic extraction based on a LVS-clean database. Therefore, you need the following components when you execute StarRC via Calibre Connectivity Interface (CCI),
  - The CDL / SPICE netlist of your design
  - The corresponding layout GDS of your design
  - ***<Calibre_LVS_runset_CCI>***, you could get from SMIC-Now website
  - ➢ **Important**

    For SMIC 40LL technology, users should apply:
    SmicSP2R_cal40_LL_sali_p1mtx_11182533_V1.14_0.lvs_for_cci
  - StarRC runsets, you could get from SMIC-Now website
- This tutorial is a general example of Calibre Connectivity Interface – StarRC flow. For advanced features and application of Calibre LVS, CCI, and StarRC, please consult

  Mentor Graphics (http://www.mentor.com/)

  Synopsys (http://www.synopsys.com/)

SMIC CONFIDENTIAL

All copyrights and IP belong to SMIC

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part I - Get Started with SMIC Calibre LVS

- **STEP 1**

  Specify the ***Environment Setting*** section in Calibre LVS runset for CCI, based on SMIC process and metallization options you preferred, e.g.

  - *TOPMETAL* — *define the total metal layers you preferred*
  - *TOP_METAL_NUM* — *define the number of top metal you preferred*
  - *TM_METAL_TYPE* — *define the type of TM you preferred*
  - ALPA_THICK — define thickness of ALPA you preferred
  - *Back_Annotation_Flow* — *define the extraction flow you preferred*

  Note : If you choose the StarRC output format as Extracted View, please set theBack_Annotation_Flow to 1, If you choose the StarRC output format as netlist, please set theBack_Annotation_Flow to 2.

- **STEP 2**

  Set the OPTION RC_RUN TRUE

- **STEP 3**

  Add the following Calibre QUERY command into the Calibre LVS rule file

  "MASK SVDB DIRECTORY <Calibre_DB_Directory> CCI QUERY "

  Note : <Calibre_DB_Directory> is the directory containing the Calibre LVS output database, default name is svdb.

SMIC CONFIDENTIAL

All copyrights and IP belong to SMIC

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part I - Get Started with SMIC Calibre LVS (*Conti.*)

- Here are the settings for metallization option 1P9M_1TMc_ALPA1 as an example,

Terminal | Terminal | Terminal

```
////////////////////////////////////////////////////////////////////////////////
//                                                                            //
//                            Environment Setting                             //
//                                                                            //
////////////////////////////////////////////////////////////////////////////////


//*Please set the following options to proper values according to your design:
//==============================================================================
//*OPTION 1: Define IO. The value can be 1825 or 1833. Default value is 1825.
//*For 40nm Logic LL 1.1/1.8/2.5V process, set this option to 1825;
//*For 40nm Logic LL 1.1/1.8/3.3V process, set this option to 1833;

#DEFINE IO 1825
//==============================================================================
//*OPTION 2a: Define total number of Metal.The value can be 10~4. Default value is 10.

#DEFINE TOPMETAL 9
//==============================================================================
//*OPTION 2b: Define type of top metal.The value can be single or double.
//*Default value is double.
//*If you use double top metal process, set this option to double(Lower Case).
//*If you use single top metal process, set this option to single(Lower Case).
//*If you set TOPMETAL to 10, you must set option TOP_METAL_NUM to double, single top
//*metal for 1P10M is not allowd!
//*If you set TOPMETAL to 4,  you must set option TOP_METAL_NUM to single, double top
//*metal for 1P4M is not allowed!

#DEFINE TOP_METAL_NUM single
//==============================================================================
//*OPTION 2c: Define top metal type.The value can be 0 or 1. Default value is 0.
//*0: TM2(122:0) is chosen as top metal2, TM2 width is 0.4um, and TM2 thickness is 9K;
//*1: MTT2(231:0) is chosen as top metal2,MTT2 width is 1.5um, and MTT2 thickness is 34k;

#DEFINE TM_METAL_TYPE 0
//==============================================================================
//*OPTION 2d: Define thickness of ALPA.The value can be thin or thick. Default value is thin.
//*THIN:  ALPA thickness is 14.5K,the model name is ralpa_2t_ckt or ralpa_3t_ckt.
//*THICK: ALPA thickness is 28K, the model name is ralpa_2p8_2t_ckt or ralpa_2p8_3t_ckt.

#DEFINE ALPA_THICK THIN
//==============================================================================
//*OPTION 3a: Define XRC RUN or not.The value can be TRUE or FALSE(Upper Case) .
//*Default value is FALSE

#DEFINE XRC_RUN TRUE
//==============================================================================
//*OPTION 3b: Define the back annotation flow.The value can be 1 or 2,Default value is 1.
//*For BJT's back annotation, XRC_RUN needs to be TRUE.
//*1: extrview flow, LVS will reserve some extra parameters for back-annotation.
//*2: netlist flow, LVS will not reserve the back-annotation parameters.

#DEFINE Back_Annotation_Flow 2
```

SMIC CONFIDENTIAL

All copyrights and IP belong to SMIC

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part I - Get Started with SMIC Calibre LVS (*Conti.*)

- An example for add QUERRY command is as follows,

Terminal

File Edit View Terminal Tabs Help

```
LAYOUT SYSTEM GDSII

#IFDEF CASE_SENSITIVITY TRUE
LAYOUT CASE YES
SOURCE CASE YES
LVS COMPARE CASE YES
#ENDIF

LVS REPORT "lvs.rep"
LVS REPORT OPTION A B C D S

UNIT CAPACITANCE FF
UNIT RESISTANCE OHM
UNIT LENGTH U

#IFDEF RC_RUN TRUE
PEX NETLIST LUMPED "lumped.dist" HSPICE LAYOUT GROUND VSS MASK
PEX REPORT LUMPED none
PEX REPORT DISTRIBUTED none
PEX NETLIST DISTRIBUTED "distributed.dist" HSPICE LAYOUT GROUND VSS MASK
//MASK SVDB DIRECTORY "svdb" XRC QUERY
MASK SVDB DIRECTORY "svdb" CCI QUERY
#ELSE
MASK SVDB DIRECTORY "svdb" QUERY
#ENDIF

LVS SPICE REPLICATE DEVICES YES //added due to solve 'RC == issue'
LVS Spice Strict WL            YES

ERC MAXIMUM RESULTS 100
ERC RESULTS DATABASE erc.db
ERC SUMMARY REPORT erc.rep
FLAG SKEW YES
FLAG OFFGRID YES
LVS ISOLATE SHORTS YES
                                                                    564,0-1        4%
```

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part I - Get Started with SMIC Calibre LVS (*Conti.*)

- **STEP 4**

  Follow the usage of Calibre LVS, specify the ***SOURCE***, ***LAYOUT*** in the deck, respectively.

Terminal

File Edit View Terminal Tabs Help

```
    I    SmicSP1RR1R_cal40_LLRF_sali_plmtx_11182533_P  Row 892  Col 1    0:13  Ctrl-K H for help

SOURCE PATH "/home/PDKQAmgr/testcase/svtinv.cdl"
SOURCE PRIMARY "svtinv"
SOURCE SYSTEM SPICE

LAYOUT PATH "/home/PDKQAmgr/testcase/svtinv.gds"
LAYOUT PRIMARY "svtinv"
LAYOUT SYSTEM GDSII

PRECISION 1000

//LAYOUT CASE YES
//SOURCE CASE YES

LVS REPORT "lvs.rep"
LVS REPORT OPTION A B C D S

UNIT CAPACITANCE FF
UNIT RESISTANCE OHM
UNIT LENGTH U

#IFDEF CCI_RUN TRUE
PEX NETLIST LUMPED "lumped.dist" HSPICE LAYOUT GROUND VSS MASK
PEX REPORT LUMPED none
PEX REPORT DISTRIBUTED none
PEX NETLIST DISTRIBUTED "distributed.dist" HSPICE LAYOUT GROUND VSS MASK
MASK SVDB DIRECTORY "svdb" CCI QUERY
#ELSE
MASK SVDB DIRECTORY "svdb" QUERY
#ENDIF
```

→ **CDL / SPICE netlist**

→ **GDS file**

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part I - Get Started with SMIC Calibre LVS (*Conti.*)

- **STEP 5**

  In order to prevent the parasitic R, C parameters of RF devices to be double extracted, SMIC provide hierarchical and hybrid flow for extraction. It needs to apply ***hcell_list*** when running Calibre LVS and then setting ***SKIP_PCELLS*** in ***CCI_cmd***. It requires to run Calibre LVS-H by the following commend syntax,

  calibre -lvs -hier -hcell ***hcell_list*** *-spice ./svdb/<top_cell>.sp* ***<Calibre_LVS_runset_CCI>***

  where ***<top_cell>*** is the top cell name of the design
  <Calibre_LVS_runset_CCI> is the Calibre LVS runset for CCI, which is provided by SMIC

➢ **Note** **This command is case-sensitive.**

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part II - Output from Calibre Query Server

- SMIC provides ***query_cmd*** in the released TAR package. By using the query_cmd, it will be easy to dump CCI output for StarRC.

- **STEP 1**

  Edit ***query_cmd***. Replace the keyword "***TOP_CELL***" by the top cell name of your design.

Terminal

File Edit View Terminal Tabs Help

```
LAYOUT NETLIST DEVICE LOCATION CENTER
GDS SEED PROPERTY ORIGINAL
RESPONSE FILE TOP_CELL.GDS_MAP
GDS MAP
RESPONSE DIRECT

GDS NETPROP NUMBER 5
GDS PLACEPROP NUMBER 6
GDS DEVPROP NUMBER 7

GDS WRITE TOP_CELL.agf

LAYOUT NETLIST TRIVIAL PINS YES
LAYOUT NETLIST EMPTY CELLS YES
LAYOUT NETLIST NAMES NONE
LAYOUT NETLIST PRIMITIVE DEVICE SUBCKTS NO
LAYOUT NETLIST PIN LOCATIONS YES
LAYOUT NETLIST HIERARCHY AGF
LAYOUT NETLIST WRITE TOP_CELL.nl

LAYOUT NAMETABLE WRITE TOP_CELL.lnn
NET XREF WRITE TOP_CELL.nxf
INSTANCE XREF WRITE TOP_CELL.ixf

CELL EXTENTS WRITE TOP_CELL.extents.txt
PORT TABLE CELLS  WRITE TOP_CELL.cells_ports

RESPONSE FILE TOP_CELL.devtab
DEVICE TABLE
RESPONSE DIRECT
                                        1,15          All
```

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part II - Output from Calibre Query Server (*Conti.*)

- **STEP 2**

  Run the following command to output database for StarRC,

  calibre -query_input query_cmd -query svdb <top_cell>

  where ***<top_cell>*** is the top cell name of the design

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part III - Get Started with SMIC StarRC

- SMIC provides a template of ***star_cmd*** for CCI in the released TAR package, which is named as ***CCI_cmd***. It is used to set options of StarRC CCI. All the options in the template are recommended. Users can modify these settings based on their own design. Here we will use the file ***CCI_cmd*** for the following demonstration.

- **STEP 1**

  MAGNIFICATION_FACTOR : 0.9

  MAGNIFY_DEVICE_PARAMS : NO

➢ **Important**

These 2 items are **essential settings** in ***CCI_cmd*** for SMIC 40LL tech node, or it will result in incorrect extraction and post-sim results by StarRC!!

- **STEP 2**

  Specify ***BLOCK*** : Input the top cell name of your design

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part II – Get Started with SMIC StarRC (*Conti.*)

- **STEP 3**

  Specify ***TCAD_GRD_FILE*** : Input the full path of the NXTGRD file. Note that the selected metallization options of the NXTGRD file MUST be consistent to your previous settings in Calibre LVS deck. For example, in the foregoing steps, we set

  DEFINE TOPMETAL 9

  DEFINE TOP_METAL_NUM single

  DEFINE TM_METAL_TYPE 0

  DEFINE ALPA_THICK THIN

  the corresponding ***TCAD_GRD_FILE*** should be the NXTGRD file for ***1P9M_8Ic_1TMc_ALPA1***

- **STEP 4**

  Specify ***MAPPING_FILE*** : Familiar to ***TCAD_GRD_FILE***, select the corresponding mapping file which matches the specified metallization options in Calibre LVS deck

SMIC CONFIDENTIAL

All copyrights and IP belong to SMIC

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part II - Get Started with SMIC StarRC (*Conti.*)

- **STEP 5**

  SKIP_PCELLS: *dnwnvar11ll_ckt* dnwnvar25ll_ckt* dnwpvar11ll_ckt* dnwpvar25ll_ckt* nvar11ll_ckt* nvar25ll_ckt* pvar11ll_ckt* pvar25ll_ckt* mom_2t_ckt* mom_3t_ckt* mom_4t_ckt* mom_5t_ckt* mom_prim2t* mom_prim* mom_prim4t* mom_prim5t**

➢ **Important**

In order to prevent the parasitic R, C parameters of RF devices to be double extracted, SMIC provide hierarchical and hybrid flow by setting ***SKIP_PCELLS*** in ***CCI_cmd***. When the devices listed in the ***SKIP_PCELLS***, StarRC will treat them as gray box during parasitic extraction. This setting is recommended, especially for RF blocking.

- **STEP 6**

  Specify ***NETLIST_FILE*** : Specify the output netlist name

- **STEP 7**

  Specify ***CALIBRE_RUNSET*** : Specify the full path of Calibre LVS runset for CCI, which is provided by SMIC

SMIC CONFIDENTIAL

All copyrights and IP belong to SMIC

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part II - Get Started with SMIC StarRC (*Conti.*)

- Overview the critical settings in ***CCI_cmd***

```
* TechReport
MAGNIFICATION_FACTOR:0.9
MAGNIFY_DEVICE_PARAMS:NO
MODE:400
NETLIST_PASSIVE_PARAMS : YES
EXTRACT_RES_BODY_COUPLING: YES
EXTRACT_VIA_CAPS:YES
BLOCK: TOP_CELL
CASE_SENSITIVE: NO
TCAD_GRD_FILE: StarRC_40LL_1P6M_5Ic_1TMc_ALPA2_TYP.nxtgrd
MAPPING_FILE: StarRC_40LL_1P6M_5Ic_1TMc_ALPA2_tran_CCI.map
EXTRACTION: RC
COUPLE_TO_GROUND: NO
POWER_EXTRACT: YES
SKIP_CELLS: !*
SKIP_PCELLS:dnwnvar11ll_ckt* dnwnvar25ll_ckt* dnwpvar11ll_ckt* dnwpvar25ll_ckt* nvar11ll_ckt* nvar25ll_ckt* pvar11ll_ckt
* pvar25ll_ckt* mom_2t_ckt* mom_3t_ckt* mom_4t_ckt* mom_5t_ckt* mom_prim2t* mom_prim* mom_prim4t* mom_prim5t*
NETLIST_FILE: XXX.spf
NETLIST_FORMAT: SPF
XREF: YES
TRANSLATE_RETAIN_BULK_LAYERS:YES
CALIBRE_RUNSET: SmicSP2R_cal40_LL_sali_p1mtx_11182533_V1.14_0.lvs_for_cci
CALIBRE_QUERY_FILE: query_cmd
CALIBRE_STRIP_LVS_REPLICATES:YES
COUPLE_TO_PCELL_PINS: YES
CALIBRE_LVS_DEVICE_TYPE_MOS: n11ll_ckt n11ll_dnw_ckt p11ll_ckt nt11ll_ckt nhvt11ll_ckt nhvt11ll_dnw_ckt phvt11ll_ckt nlv
t11ll_ckt nulvt11ll_ckt nulvt11ll_dnw_ckt nlvt11ll_dnw_ckt plvt11ll_ckt pulvt11ll_ckt n25ll_ckt n25ll_dnw_ckt p25ll_ckt
nt25ll_ckt nod33ll_ckt nod33ll_dnw_ckt pod33ll_ckt nud18ll_ckt nud18ll_dnw_ckt pud18ll_ckt ntod33ll_ckt ntud18ll_ckt nld
50ll_ckt pld50ll_ckt nld50llod_ckt pld50llod_ckt
CALIBRE_LVS_DEVICE_TYPE_RES:rndif_3t_ckt rpdif_3t_ckt rnwsti_3t_ckt rnwaa_3t_ckt rndifsab_3t_ckt rpdifsab_3t_ckt rnpo_3t
_ckt rppo_3t_ckt rnposab_3t_ckt rpposab_3t_ckt rhrpo_3t_ckt rndif_2t_ckt rpdif_2t_ckt rnwsti_2t_ckt rnwaa_2t_ckt rndifsa
b_2t_ckt rpdifsab_2t_ckt rnpo_2t_ckt rppo_2t_ckt rnposab_2t_ckt rpposab_2t_ckt rhrpo_2t_ckt rm1_2t_ckt rm1_3t_ckt rm2_2t
_ckt rm2_3t_ckt rm3_2t_ckt rm3_3t_ckt ralpa_2p8_2t_ckt ralpa_2p8_3t_ckt rm4_2t_ckt rm4_3t_ckt rm5_2t_ckt rm5_3t_ckt rtm2
_2t_ckt rtm2_3t_ckt
```

Essential settings for 40LL tech node

The selected GRD file and MAP files MUST be consistent to the metallization settings of LVS.

Blocking devices during extraction

Specify Calibre LVS runset for CCI

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14

Part II - Get Started with SMIC StarRC (*Conti.*)

- **STEP 8**

  Run StarRC by the following command,

  StarXtract -clean CCI_cmd

➢ **Note** **This command is case-sensitive.**

SMIC CONFIDENTIAL

According to:40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow;
Attachment No.:TD-LO40-XS-2037-quick-start_on_smic_cci-starrc;Rev.:0 2016-06-14